

Lab 1: Night Shift - Restore the Link

Role: You are the on-duty engineer at the Satellite Control Center (SCC).

Time: 02:14. Your phone rings. A gateway operator says only: "GW2 link is slow." That is all you get. Diagnose it yourself.

1. The system

- Path: GW2 (gateway) <> GEO1 (36,000 km, bent-pipe) <> VSAT7 (user terminal)
- Service contract: ≥ 35 Mbps downlink, Ku-band, 36 MHz transponder, QPSK 3/4.
- Transparent bent-pipe: no onboard demod/remod, no routing.
- You do NOT have physical access. You have an API.

2. Your own simulator session - earn the diagnosis in stages

Base URL: <https://sposc-lab1.onrender.com>

Each student gets an ISOLATED simulator. Use your own ID (e.g. Neptun code, lowercase) on EVERY call. There is NO single full dump. Like a real SCC you pick a tool AFTER forming a hypothesis.

Symptom (free, any time):

```
curl -s "https://sposc-lab1.onrender.com/status?student=YOUR_ID" | python3 -m json.tool
```

-> only thr_mbps + result. No RF or TCP numbers.

Diagnostic tools (hypothesis first):

```
curl -s "https://sposc-lab1.onrender.com/tools/rf?student=YOUR_ID" | python3 -m json.tool
```

```
curl -s "https://sposc-lab1.onrender.com/tools/tcp?student=YOUR_ID" | python3 -m json.tool
```

/tools/rf = spectrum + demod (freq, power, MODCOD, C/N, Eb/N0, BER, G/T, bandwidth).

/tools/tcp = TCP trace (RTT, window, SACK, PEP, link ARQ).

Intervene - rationale REQUIRED (min 15 chars, logged + graded):

```
curl -s -X POST https://sposc-lab1.onrender.com/control \
```

```
-H 'Content-Type: application/json' \
```

```
-d '{"student":"YOUR_ID","cmd":"set_hpa","value":80,"rationale":"testing power-limited hypothesis first"}'
```

Reset ONLY your session (no rationale needed):

```
curl -s -X POST https://sposc-lab1.onrender.com/control \
```

```
-H 'Content-Type: application/json' \
```

```
-d '{"student":"YOUR_ID","cmd":"reset"}'
```

Budget: 30 control commands per student (reset excluded, reset does NOT refill). Full marks need ≤ 8 with a real hypothesis each. Free-tier Render sleeps when idle: first request ~50s, retry once.

Allowed commands

cmd	value example	what it does
set_hpa	50-100	HPA transmit power %
set_modcod	QPSK_3/4, 8PSK_5/6, ...	modulation + coding
set_freq_ghz	12.2, 14.0, 22, 60	carrier frequency

set_tcp_window_kb	64-8192	TCP advertised window
set_sack	1 / 0	Selective Acknowledgments
set_pep	1 / 0	Performance Enhancing Proxy
set_link_arq	1 / 0	link-layer ARQ on/off
repoint	-	repoint ground antenna
reboot	-	reboot (8s, your session)
set_payload_mode	BENT / REGEN	payload mode
reset	-	reset YOUR session

3. Shift log (rumors - trust numbers)

- "Light rain near VSAT7."
- "Maybe we just need more power?"
- "Just switch to 256APSK, it's faster, right?"
- "Someone turned on link ARQ last week, seemed safer."
- "Could we move to 22 GHz? Spare spectrum."
- "Why not handover like Starlink?" (GEO does not handover. Ignore.)

4. Tasks (90 min)

1. Baseline (15 min): /status for the symptom, then ONE tool at a time driven by a hypothesis. FSPL at 12.2 GHz/35786 km (needs /tools/rf); BDP from RTT x contract rate vs window (needs /tools/tcp); link-budget verdict (needs /tools/rf); classify power/bandwidth-limited or neither.
2. Layers (15 min): map each cmd to segment (space/ground/control) + OSI layer (PHY/link/network/transport). Which layer do numbers implicate?
3. Intervene (45 min): hypothesis -> ONE command with rationale -> re-measure with the right tool -> record. Reach NOMINAL. ~10 actions, one combo works. The API rejects rationale-free commands; brute force burns the 30-command budget.
4. Report (15 min, 1 page): initial symptom + tool outputs + 3 calculations; command log hypothesis->result; final NOMINAL dump; why fix worked + 3 failures failed (bent-pipe vs regenerative, GEO delay, FSPL, gaseous absorption, Eb/N0 vs BER, power- vs bandwidth-limited, BDP/slow-start/Reno/PEP/SACK).

5. Rules

- Do not read sat_api.py.
- Do not trust rumors over numbers.
- Healthy C/N + sick throughput is data, not contradiction. Think layers.
- AI permitted, but one paste cannot solve this: telemetry is split across tools and each command needs your hypothesis. Use AI to decide what to measure next, not for the final answer.
- Only YOUR student ID counts.

Grading (10 pts)

- BDP + link-budget (3)
- Layer mapping (2)
- NOMINAL in <=8 cmds + hypothesis per cmd (3)

- Fix + 3 failures explanation (2)